Term Project Instructions

Sept. 9, 2026

Assignment

To help you gain experience working with real-world data sets, this project will require that you mine actual bank transaction data to solve a problem of interest to your client, a large Italian bank seeking to reduce its losses from bad loans. You will be working in teams of 3-4 students. Please ensure that all students participate equally.

This project will require that you mine actual transaction data to solve a problem of interest to your client, Banca Massiccia. a large Italian bank that is interested in understanding how to reduce the default rate on individual loans in its portfolio.

In particular, the firm wishes to get better predictions of the one-year probability of default (PD) for a prospective borrower.[1]

Context Overview

Imagine that you have been approached by Banca Massiccia, a large Italian bank. Banca Massiccia has been making loans to businesses for many years, and has experimented with different approaches to underwriting these loans, including statistical models of default. The Head of Loan Origination has asked you to assist the firm in optimizing its underwriting using the power of machine learning. The goal is to produce estimates of the probability of default (PD) for prospective borrowers so the bank can use risk-based pricing to set interest rates and underwriting fees for borrowers. One way to do this is to leverage the data that Banca Massiccia has accumulated over the past several years.

The basics of the transaction flow are as follows:

1. A potential borrower applies for a loan by providing information about the finances of the firm in the form of financial statement data.
2. The loan officer assigned to the borrower analyzes the financial data using a number of human and automated approaches to try to determine how likely it is that the borrower will default over the next year.
3. The bank officer then determines what the appropriate interest rate and underwriting fees would be for the loan to be profitable, in consideration of the default probability.
4. If the bank officer concludes that the loan may be made at a rate that is within the bank's guidelines, the loan is made, and the customer's information is transferred to the loan monitoring group, which continues to monitor both changes in the financial status of the borrower, and the payment status of the loan.
5. If the borrower defaults before the loan is repaid, the borrower's case is transferred to the "work-out" group to try to recover the shortfall through various legal remedies.

Specific Business need: Predicting Probability of Default

Banca Massiccia would like to be able to better predict the probability that a potential borrower will default on a principal or interest payment for a prospective loan over the next 12 months.

The bank has asked you to design the data mining task, mine the data, produce a model the can use, and document your results. You will need to research existing solutions to the problem.

For purposes of the Term Project, you can think of your stakeholder(s) as internal members of the bank's lending group and risk management group (in other contexts, the stakeholder might be a consulting client or a VC/incubator who may fund

[1] This data and problem are from real-world Italian companies. The data is real-world data. However, some of the details of the data have been modified for confidentiality.

you). You will need both to inform the stakeholder about your project and results and provide context by reporting on what has been done to date (elsewhere) on the problem.

It is most important to develop your models and analysis study well (within the scope of what we've discussed in class) and with a good understanding of the business problem.

You should use the frameworks and methodologies we used this semester to structure your project and writeup. Keep in mind that it may be ineffective simply to proceed linearly through the steps, and this may need to be reflected in your analysis. You can interact with me and your TA from the preparation of your initial ideas through your write-up, as a consulting group would interact with a stakeholder or funding source in preparing a research report. However, given the class size, and depending on the number of groups, there may not be as many opportunities to do so as you (or I) would like. Use your imagination, prior experience, or ask us to help fill in any gaps in your understanding.

Deliverables

#1: By **Sept 23rd** you will submit your choices for teams for projects to your TA.
You should self-organize your teams, but ask us if you need help. Please include a paragraph or two about what you are thinking you might do. There is also a section of Ed Discussion called "Students seeking teams" that we set up to help you find teams and/or members.

#2: By **Oct 5th** you will submit a **proposal** for your project with as much detail as possible your ideas. This is where you describe your detailed ideas on how you will formulate the problem.
Review your notes on problem formulation before you start this. Be sure to include: the exact (business) problem; the use scenario, the related data mining problem (and whether it is supervised or unsupervised; the unit of analysis; potential target variables (if supervised); potential features; how the results solve the business problem? etc., as well as why you are approaching this finance problem through your formulation and what finance characteristics make your formulation valuable. Similarly, please describe how you will formulate that validation of your model.

Feedback will be provided by Roger during office hours which each team will schedule independently.

#3: By **Oct 27th** you will submit a status report ("Project Update") including preliminary & results, validation, hypotheses, and any
issues that you have run into.

#4: By **Nov 16th**, you will submit your final write-up and code.

Your write-up:

1. should include the information detailed in the next section, in approximately the order given.
2. be about 15 – 25, fairly dense (but readable) slides, including any appendices you would like to include. Provide clear citations and bibliography for the external sources you use. More detail is given below.
3. An appendix detailing the contributions of each team member. All group members should contribute to the analysis and write-up is also required.

Your code will include three main modules:

1. The code you used to estimate your solution (with in-line comments and documentation where necessary);
2. The code that you want us to use to make new predictions using your model, but which does not use the holdout data to estimate any parameters; and
3. A "harness" function that will allow us to upload new data into your fitted solution (in 2, above) by calling a single function. Your harness should produce a list of results that we can use for validation.

Details on these components are given in Appendix A.

Details of the data that you will be using are given in Appendix B.

At the end of the semester we will conduct a tournament to evaluate the relative performance of your model as well, as a bank might do when comparing your proposed solution to alternatives that may be available to decide whether to use it.

Appendix A: Project Deliverables and Evaluation

Holdout-sample evaluation

In addition to reviewing your writeup, I have reserved a portion of the data in a vault at an undisclosed location. We will use this data set to evaluate your model's true out-of-sample performance using the harness code that you provide.

You may assume that the record structure of the holdout sample will be the same as that of the full training data sample, unless you are instructed otherwise. Be aware that there may be borrowers in the hold-out sample that were not in the training data.

Your model will be evaluated in terms of the accuracy of the probabilities it produces and in terms of its discriminatory power. We will also examine model robustness and face validity (i.e., do the variables and parameters of the model make sense). For purposes of the horserace, we will be examining the pairwise statistical and practical differences in the metrics we calculate, i.e., we will compare all possible tournaments of model pairs to assess performance differences using the evaluation criteria we will learn over the semester.

Output specifications:

- that your model should produce an estimate of the probability of default (PD) for each new record, i.e. one, and only one, probability for each record. You must return these predictions in a n x 1 vector, where n is the number of rows in the hold-out sample. **YOU MUST PRODUCE A PROBABILITY FOR EVERY RECORD IN THE HOLDOUT DATA.** (Note that 1/0 is technically a PD.)
- Your model should not depend in any way on the data in the hold-out sample.
- Remember that you will only be allowed to use data that you would have access to in real life at the time of your prediction for the firm in the prediction year.
- Be sure not to use label vars (e.g., '`HQ_city`') as numerical values.
- Please (please, please!) look at what your model is doing in detail before submitting it.

Pitch Deck

Your pitch deck should be something that you would use to explain your work to a client or colleague and should address both the positive aspects of your solution, and its quantitative and economic limitations, etc. Your pitch deck can be any length you like (typical submissions contain 15-25 slides with fairly dense, but readable, information content), and may include one or more technical appendices you think are helpful.

You may wish to include the information detailed below. Your write-up need not have corresponding sections or bullet points, but we should be able to find the information without searching too hard. Be as precise/specific as you can.

Business Understanding (take this seriously)

- Identify, define, and motivate the business problem.
- How (precisely) will a data mining solution address the business problem?
- What has been done in the past both from a finance and a machine learning perspective?

> **Example of how to think about reviewing previous work for stakeholders**: For this project, we are working on loan data from a large Italian bank. One way to think about how to present previous work, is to imagine that you are pitching your idea to a principal at a VC firm that has experience funding a large number of machine-learning-based FinTech startups. The VC wants you to explain how your approach is different from what the firm already knows and what other modelers might do.
>
> You would want to discuss:
> - any good ideas that you incorporated from earlier work;
> - how your default probability estimation approach is different from current approaches; and
> - why your approach is better (uniformly or in specific settings only) for loan underwriting.
>
> Don't fall into the trap of only reviewing machine learning research that has been done on this problem. Much of it is not really informed by finance. You will benefit by getting familiar with both the finance and machine learning literature on this problem.

Problem formulation including what has been done in the past and why your approach is better (uniformly or in specific settings only):

- Did you set this problem up to make it easier for your algorithm to learn the problem?
- How did you incorporate financial intuition and theory into your model?
- Why will this formulation be more successful than if you had just mined the data without a finance context?

Data Understanding including which data you used and what was problematic or particularly useful about it, as well as biases & limitations.

- Identify and describe the data that you used to address the business problem. Include those aspects of the data that we routinely talk about in class and/or in the home works, including variable definitions, calculated or derived variables, and, of course, potential biases.

Data Preparation including how you have filtered, cleaned, preprocessed and munged the data. Discuss limitations of your data handling.

- Specify how these data are integrated to produce the format required for data mining.
- Describe any preprocessing you did on individual features or variables.
  (NB: data preparation can be time consuming! Get started early.)

Modeling including a description of the approaches you explored and the formulations you used in setting up the problem to let the algorithms you used learn efficiently in a finance-aware context. Discuss which variables you selected (and why/how). This is also a good place to discuss the model itself and any insights the team got from diagnosing and examining your models' parameters and variables:

- Specify the type of models you built and/or the patterns you mined.
- Discuss choices for data mining algorithm: what are alternatives, and what are the pros and cons?
- Describe the final set of variables used in your final model.
- Discuss the *economic intuition* for your model and results.
- Discuss why & how this model should "solve" the business problem (i.e., improve along some dimension of interest to Bank Massiccia.
- Make sure to actually include the full specification of your final model!

Evaluation including how you set up tests and evaluated your solution, and how it compared to other approaches, etc. Include the specific results of your evaluation, benchmarks, etc.

- Discuss how the result of the data mining is/should be evaluated and how you evaluated your model.
- Discuss any benchmarks you used to evaluate relative performance and provide performance analysis.
- Suggest how a business case should be developed to project expected improvement.
- If this is impossible/very difficult, explain why and identify any viable alternatives.

Deployment

- Discuss how the result of your modeling should be deployed.
- Discuss issues the bank should be aware of regarding deployment:
  - Are there important ethical or legal considerations?
  - Describe settings in which your results should <u>not</u> be applied.
  - Identify the risks associated with your proposed plan and how you would mitigate them.
  - Etc.

In addition to the materials on the course website, you will probably find a number of useful sources in the library as well. These can be invaluable. Please make sure to cite any literature and sources you used in your work, and provide a bibliography referencing the source.

All team members should contribute to the analysis & write-up. Make sure to include an appendix that describes each team member's contribution to the work. Remember, the team may be asked to actually pitch your solution to the class. Make sure you have what you need inside the pitch deck to do so.

Where useful, include model output, data visualizations, etc. to explain what you did and found, the structure of your model, etc.

Include a detailed representation of your final model (e.g., regression output, a tree, etc.), & be prepared to discuss the economic intuition for it.

Model codebase

Your codebase will include three main modules implemented in Python:[2]

1. Your estimation code: The code you used to estimate your solution (with in-line comments and documentation where necessary);
2. Your prediction code: The code that you want us to use to make new predictions using your model, but which does not use the holdout data to estimate any parameters; and
3. Your harness code: A "harness" function that will allow us to upload new data into your fitted solution (in 2, above) by calling a single function. Your harness should produce a list of results that we can use for validation.

Your estimation code may use any algorithms you have studied, except for LLMs. Feed-forward ANNs are permitted.

Your prediction code may take in one or more rows of data, and produce output in a form that you find useful, *provided it does not use the any of the test data, besides the current record you are predicting, for any purpose* (e.g., you cannot calculate the mean of a variable in the test data to scale that variable in a new record). You may perform any legitimate data transformation and normalization on a data record before running your model on it, and you may post-process your model output using any legitimate operations you find useful.

Your harness code should run with a single command in our environment hosted on reasonably configured hardware without throwing any errors or halting unexpectedly. The harness should not require a user to have any knowledge of your model or code to run properly.

Make sure that any parameters you need to use (e.g., mean and SD for standardization) are either hardcoded, or in a parameter file that you provide. If you use the training data dynamically (e.g., for KNN), please make sure to provide a data file in the form that you need for your harness. Make sure that any preprocessing logic (e.g., transformations, missing value treatment, creation of dummies, etc.) is done automatically by the harness.

Your harness should not re-estimate your model at run time. Note that if you wish to use external data fields that are not in the training/holdout data, please:

- check with us at least several weeks before the due date to make sure your use of these variables is both practical and compliant; and
- make sure that your harness fetches and handles these properly without peeking into the future, and without the need for us to do anything special.

The code for the harness and the various components of the model, etc. should be well documented.

We will not debug your code if it does not run. Be certain that your model runs in your harness! Although it is not required, you may submit a test version of your harness and model to the TA one week before the project is due so that you can determine whether it will run properly on the data we will be using. Any model that cannot run on our data without modification will be disqualified from the tournament and will not be eligible for the bonus point prize. However, even if your model does not run, we will still evaluate and grade the other aspects of your project.

Presentation:

Depending on the number of teams, k projects will be selected and the corresponding project teams will pitch their solution to the class. Note that in most cases, $k<n$, where k *and* n are the number of teams presenting and total number of teams, respectively.

[2] You may use R or Python to conduct your research, EDA, etc. However, if you plan to submit your final code in R, please let us know at the time you submit your proposal. If you do not, only Python code will be accepted.

- **You will get the most out of the project if you interact with us during the development of your ideas.**
- **Please feel free to come talk to us about your ideas as time permits.**

Appendix B: Data description

Data Overview

The IT department at Banca Massiccia has agreed to provide you with information on previous borrowers, including information about the company type, industry, etc. as well as financial statement data on an annual basis for each borrower. Because the firm is located in the EU, where privacy and data sharing are heavily regulated, some of the data that Banca Massiccia collects may not be shared with your team. The Data Dictionary section, below, provides the definitions of the variables in the training data set.

Data Conventions

The data in both the training and holdout samples will have identical structures and will conform with the following conventions:

- Each row is one firm-year
- Observations are given at annual intervals.
- There will be 44 variables in total.
- All quantities (except ratios) are reported in €.
- Only firms with > €1.5MM in assets are included in the data.
- Only non-finance/insurance firms are included in the data.
- There is no information in the training data regarding defaults that occurred after 12/31/2012
- The value for the default date (`def_date`) is `NA` in the holdout sample

The holdout sample is drawn from a future time period and will include both firms that are in the training data and those that are not.

The training and holdout samples will have identical record structures and will conform with the following conventions the data dictionary given on the next page, unless you are informed otherwise.

`train.csv` (financial statements and default dates behavior)

Variable name (column name, feature, etc.)	**Description**
id	Firm identifier
HQ_city	City of main branch
legal_struct	Legal structure of the firm
ateco_sector	Industry sector code (see ATECO sector definition doc)
fs_year	Year of the financial statement
asst_intang_fixed	Intangible assets
asst_tang_fixed	Tangible Assets
asst_fixed_fin	Financial assets
asst_current	Current assets
AR	Accounts receivable
cash_and_equiv	Cash & equivalent holdings
asst_tot	Total assets
eqyt_tot	Total equity
eqty_corp_family_tot	Total equity for entire group ("family")
liab_lt	Long-term liabilities
liab_lt_emp	Long-term liab to employees
debt_bank_st	Short-term bank debt
debt_bank_lt	Long-term bank debt
debt_fin_st	Short-term debt other
debt_fin_lt	Long-term debt other
AP_st	Short-term accounts payable
AP_lt	Long-term accounts payable
debt_st	Short-term debt
debt_lt	Long-term debt other
rev_operations	Operating revenue
COGS	COGS (Cost of goods sold)
prof_operating	Operating profit
goodwill	Goodwill
inc_financing	Financial income
exp_financing	Financial expenses
prof_financing	Financial profit
inc_extraord	Extraordinary income
taxes	Taxes
profit	Net profit
days_rec	Days recievables
ebitda	Earnings before interest, taxes, depreciation, and amortization
roa	Return on assets
roe	Return on equity
wc_net	Net working capital
margin_fin	(Equity - Fixed assets)
cf_operations	Operating cashflow